Building RMIT's Clinical Data Repository

Student Capstone Kick-off | Powered by Databricks

The Challenge at RMIT R&D

Modern clinical and health research relies on analysing vast amounts of multi-modal data, but researchers face critical bottlenecks:

- **Fragmented Silos:** Data is trapped across disparate EHR systems, trial databases, and unstructured medical notes.
- **Complex Extraction:** Researchers spend excessive time navigating systems and harmonising data manually.
- **Strict Compliance:** Securing Protected Health Information (PHI) and ensuring compliance with the Australian Privacy Principles is difficult across disconnected tools.



The Capstone Objective

Architect and build a next-generation Clinical Data Repository (CDR) on the Databricks Data Intelligence Platform to solve RMIT's data challenges.



Unified Lakehouse

Automate the ingestion of complex clinical datasets into a single, scalable environment using the Medallion Architecture.



Strict Governance

Enforce strict, centralised data governance and secure sensitive PHI using Unity Catalog access controls.



Democratised AI

Empower researchers to interrogate data and generate insights using intuitive, natural language via AI/BI Genie.

Core Deliverables: Platform & Governance

1. Data Platform Prototype

A functional implementation demonstrating automated ingestion, quality checks, and transformation of clinical data.

- Implement Medallion Architecture (Bronze, Silver, Gold layers).
- Build scalable ETL/ELT pipelines using PySpark and SQL.
- Utilise Delta Lake for reliable and performant storage.

2. Robust Governance

A working configuration proving how highly sensitive clinical data will be actively safeguarded and audited.

- Configure Databricks Unity Catalog for unified governance.
- Demonstrate Role-Based Access Control (RBAC).
- Implement data masking, row-level, and column-level security for PII/PHI.

3. Applied AI/BI Showcase

Democratising Clinical Insights

You will implement a functional AI/BI Genie Space configured on top of your "Gold" layer data.

This demonstration will show how an RMIT researcher—without SQL expertise—can use natural language prompts to instantly generate accurate data summaries and visual dashboards.

> *"Show me the readmission rates for cardiovascular patients over the last 6 months broken down by age group."*



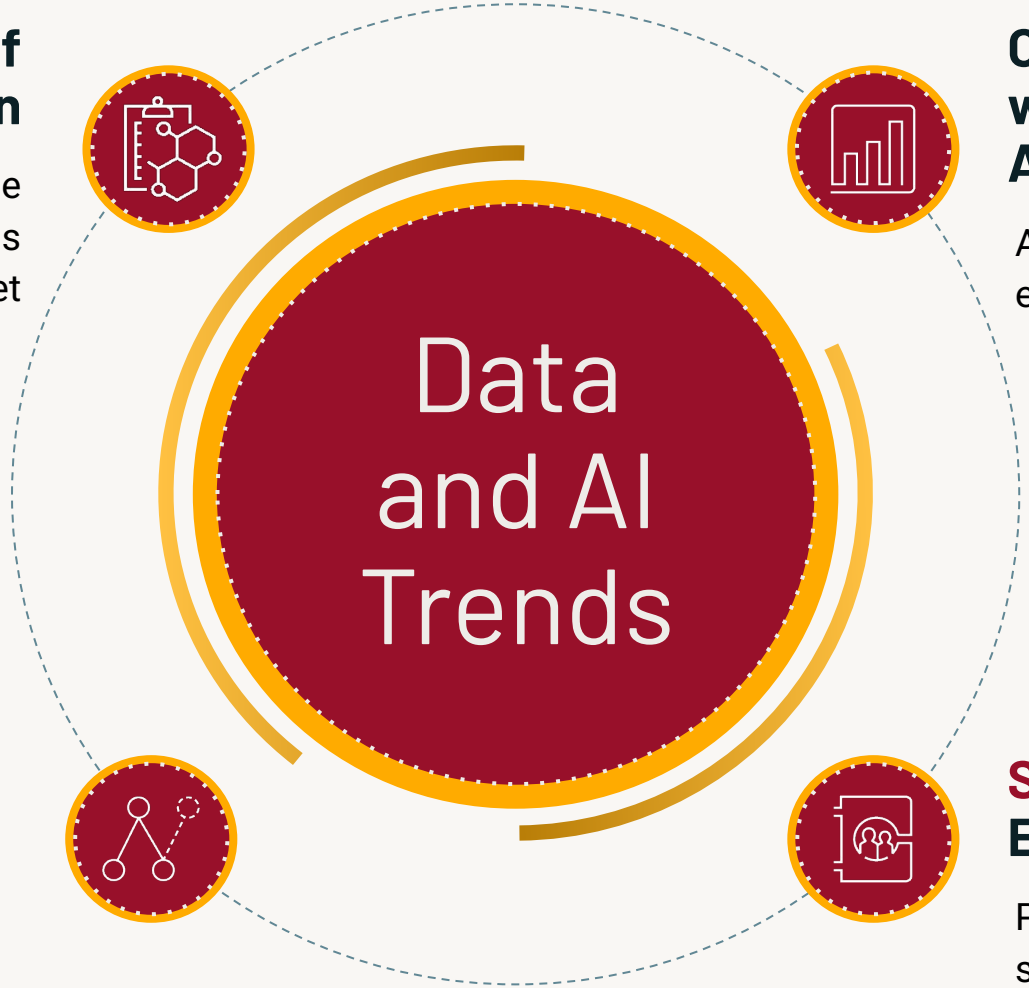
Required Skill Set for Success

- **Programming & Querying:** Strong proficiency in Python for data manipulation and SQL for structuring data and setting up semantic layers.
- **Data Engineering (ETL/ELT):** Solid understanding of data pipelines, Lakehouse concepts, and relational data modelling.
- **Data Governance Fundamentals:** Conceptual understanding of privacy, Access Control Lists (ACLs), RBAC, and handling PII/PHI safely.
- **Cloud Infrastructure:** A foundational understanding of modern cloud computing environments (AWS, Azure, or GCP).
- **AI & Semantic Engineering:** Understanding of how LLMs interpret data, prompt structuring, and providing context via a semantic layer to help AI/BI tools.

Key Data and AI Trends in R&D

AI-Driven Acceleration of Drug Discovery and Design

AI and ML enables reducing the time and cost required to bring new drugs to market



Optimization of Clinical Trials with AI and Real-Time Analytics

AI is revolutionizing the design and execution of clinical trials

Data-Driven Decision Making and Predictive Analytics

The integration of big data and AI is enabling pharmaceutical companies to make more informed, data-driven decisions across the R&D pipeline

Strategic Partnerships and Ecosystem Collaboration

Pharmaceutical organizations are forming strategic partnerships with technology companies, AI startups, and academic institutions

Insights from Our Customers

Challenges we hear from Life Sciences customers

1

AI Urgency

But **no real path beyond** a chatbot

2

Modernization Under Cost Pressure

Conflicting needs to modernize and save costs

3

Underestimated complexity

Pocketed use cases and **difficulties to challenge** bigger systems

Based on use-cases from 14 of the 15 largest global Pharma

How Databricks addresses these challenges

End to end Data and AI Platform

1

Data and AI

Complete agent systems platform with **Data and AI intelligence**

2

Costs and Performance

Reduced TCO to enable other investments, with features like **Lakebase** for further optimization

3

Success at Scale

Fast and scalable data and AI adoption with end-to-end platform

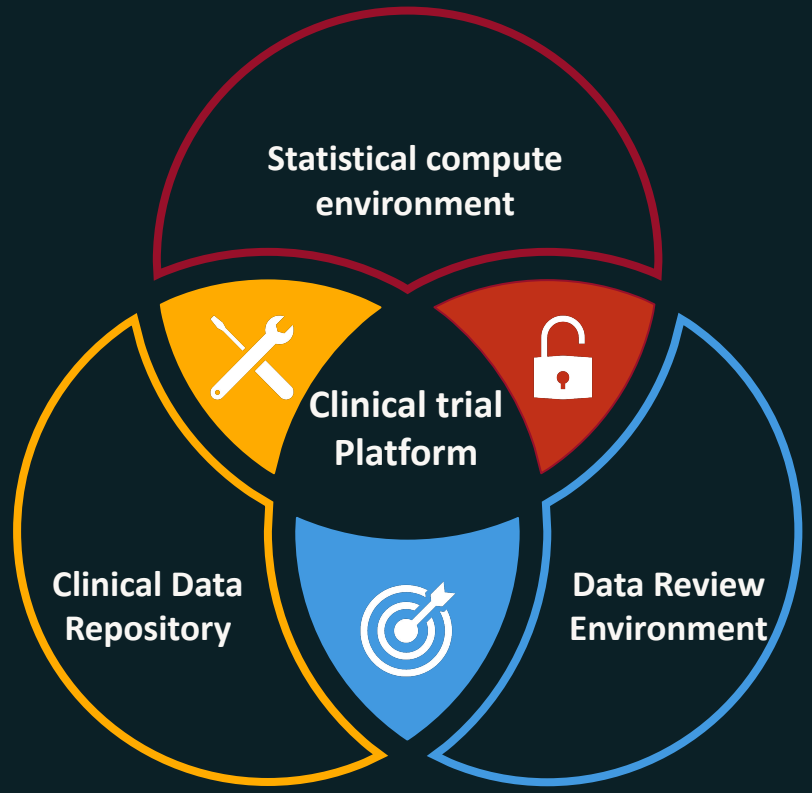
Next Generation of Clinical Data management

Clinical Data Repository

- Real-time data processing
- New data modalities - data from wearables and IoT devices, images and video/audio data as part of clinical trials
- Decentralized clinical trials
- Simplification of data collection and integrations

AI and Automation at Scale

- AI- suggested Mappings and Checks
- Custom models and reusability of clinical data
- AI support on protocol generation and protocol amendment propagation
- AI support on compliance by change document generation



Statistical Compute Environment

- Reducing dependencies on SAS, moving to R and Python as primary languages
- Open Source Packages and ecosystem, unified data models

Data Review Environment

- Real-time data processing and interactive dashboards, empowering trial managers with actionable insights and faster decision-making
- New custom apps for complex visualizations
- Natural language interface to perform data review

Databricks for Clinical Trials

Key Highlights

Clinical Data Platform

User Experience

- Self Service Facility
- Higher Productivity
- Increased Cross Functional Collaboration and Transparency

Future Scale

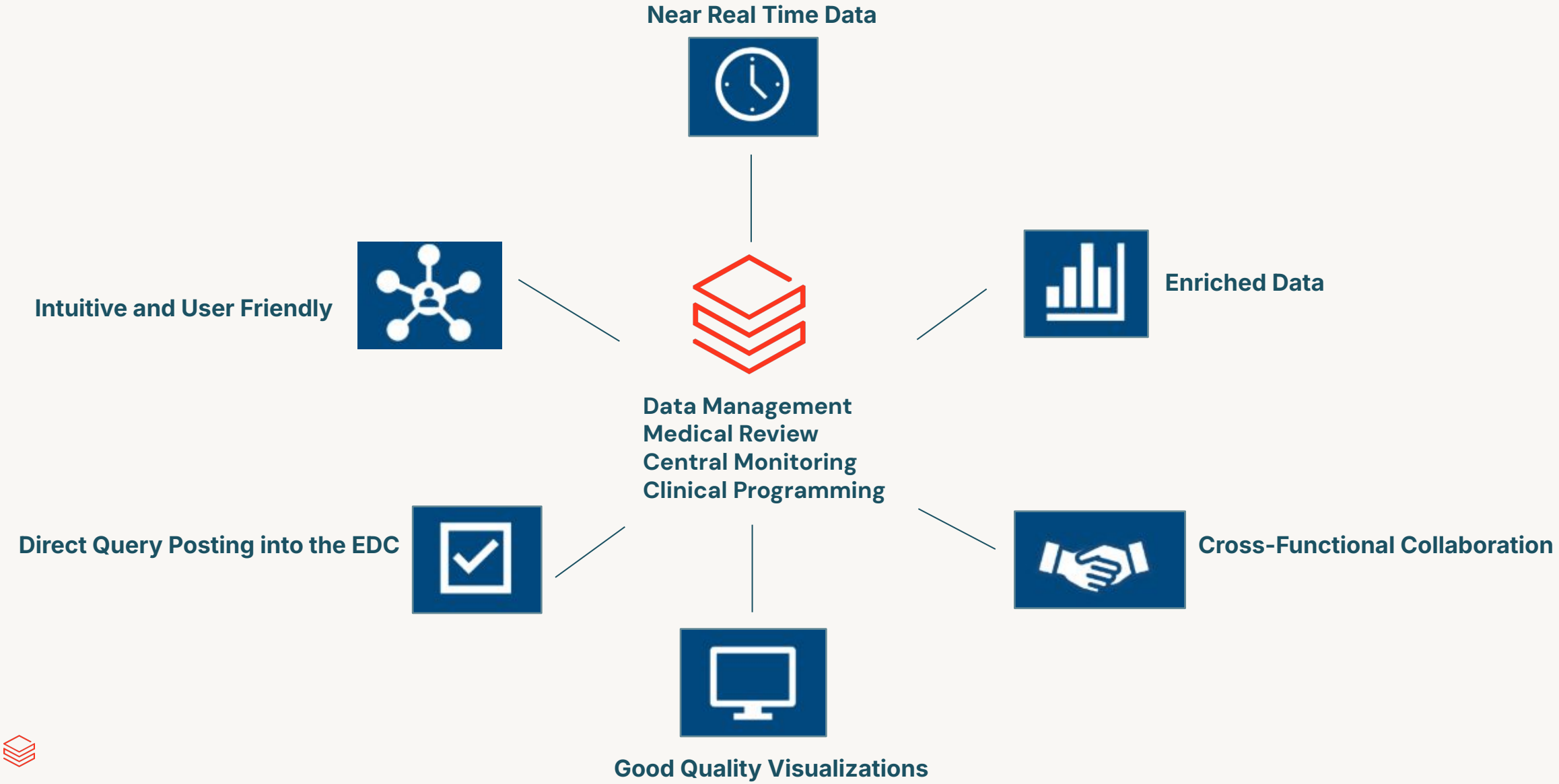
- Improved Decision Making
- Introduction of AI/ML enhance Data Review
- Higher Dataflow

Quality & Financial Value

- Reduced Cycle Time creates Revenue Uplift
- Better Data Availability & Improved Risk Mngmt.
- Reduced effort leading to operational saving

Databricks Clinical Data Platform

Benefits – One System for all



Where Databricks can help in drug discovery lifecycle



- AI for drug target prioritization (deep learning, CNN)
- Drug molecule discovery
- Genomics
- Deep Learning to Drive Novel Drug Target Identification
- AI Platforms Analyze Complex Biological Data from Different Sources to Arrive at Suitable Drug Targets

AI in drug design, predicting 3D structures of target

AI to determine drug activity

AI in chemical synthesis

AI in drug repurposing

AI in drug screening (prediction of toxicity, bioactivity, physicochemical property)

AI in molecule optimisation (computer vision, image data processing)

- Site selection (predictive analytics, DS/ML to identify target sites)
- DS for preclinical trial optimisation
- Clinical trial prediction

- Clinical data repository
- In-trial analytics and DS to streamline conversion to SDTM and validation checks
- Post-trial analytics
- Unified clinical review environment
- Simplifying submission process
- Identifying the most likely requests from government regulators and incorporating the answers from the get-go

- Data sharing with CROs, contract development and manufacturing companies
- Faster decisions on a medicine's potential
- Bring to market DS, marketing reporting and DS/ML
- Reporting on drug distribution
- NLP for regulatory documents base (documents processing, search)

How data flows in clinical trial process



Components of Clinical Data Repository

- **Study information repository** – enables creation setup and maintenance of studies
- **Metadata repository** – repository that allows to quickly setup study based on the standards, it also allows to build/define mappings and transformation to conform data in accordance with defined standards
- **Clinical Data Lake** – global secure GxP compliant repository that stores all structured and unstructured clinical data and documents
- **Information quality** – auditing and reporting component to guarantee traceability, reproducibility and compliance of clinical data transformations
- **Data standardisation packages** – automated libraries (powered by AI) to support with conversion to standards (SDTM, ADaM), submission ready packages
- **Orchestration system** – provides capability to schedule data processing and also overview of overall data processing flow



Why companies want to modernise CDR platforms

Challenges and missing functionality from exiting CDR solutions

1. **Monolithic Architectures** (little flexibility in terms of performance, scalability, and costs), monolithic storage (limited data storage lifecycle)
2. **Limited support of unstructured data**, schema evolutions, data versioning (multiple copies of data for backwards compatibility, archiving purposes, different stages of the process, etc.)
3. **Limited support of AI/ML**
4. For SAS-only solutions, limited possibility of connected other analytical tools, limited support of 3d party apps and non-SAS files
5. **Complicated security model** (with large number of user groups, limited support of SSO, limited traceability), limited compliance automation e.g. access mgmt and audit readiness data flow automation from ingestion to consumption
6. **Low flexibility of data management tools**, e.g., quality check framework, governance tools , data review framework
7. **Limited monitoring and alerting**, both from a system and a data perspective (lack of monitoring dashboards with visual analytics on events (e.g. data transfers), system usage, performance, dataflow, query usage and execution etc., notification of data consumers about data load)
8. **Complicated or limited event-base data management** (f.e. to automatically return data transfer findings/errors to external partners)
9. **Linear processing and lack of parallelization** to speed up the process
10. **Lack of built in Data Provenance and lineage**, no Integrated Governance tooling, schema Management, orchestration
11. **Complicated integration with other tools**



How Databricks supports CDR use cases

Flexible Architecture
Based on the Delta lake and utilizing cloud storage allows flexibility in performance and costs

Full support of DS/ML and AI
With tools like MLFLow, AutoML etc, Databricks supports full end-to-end DS/ML lifecycle

Delta lake
Support structured and unstructured data, schema evolution, versioning, time travel and more

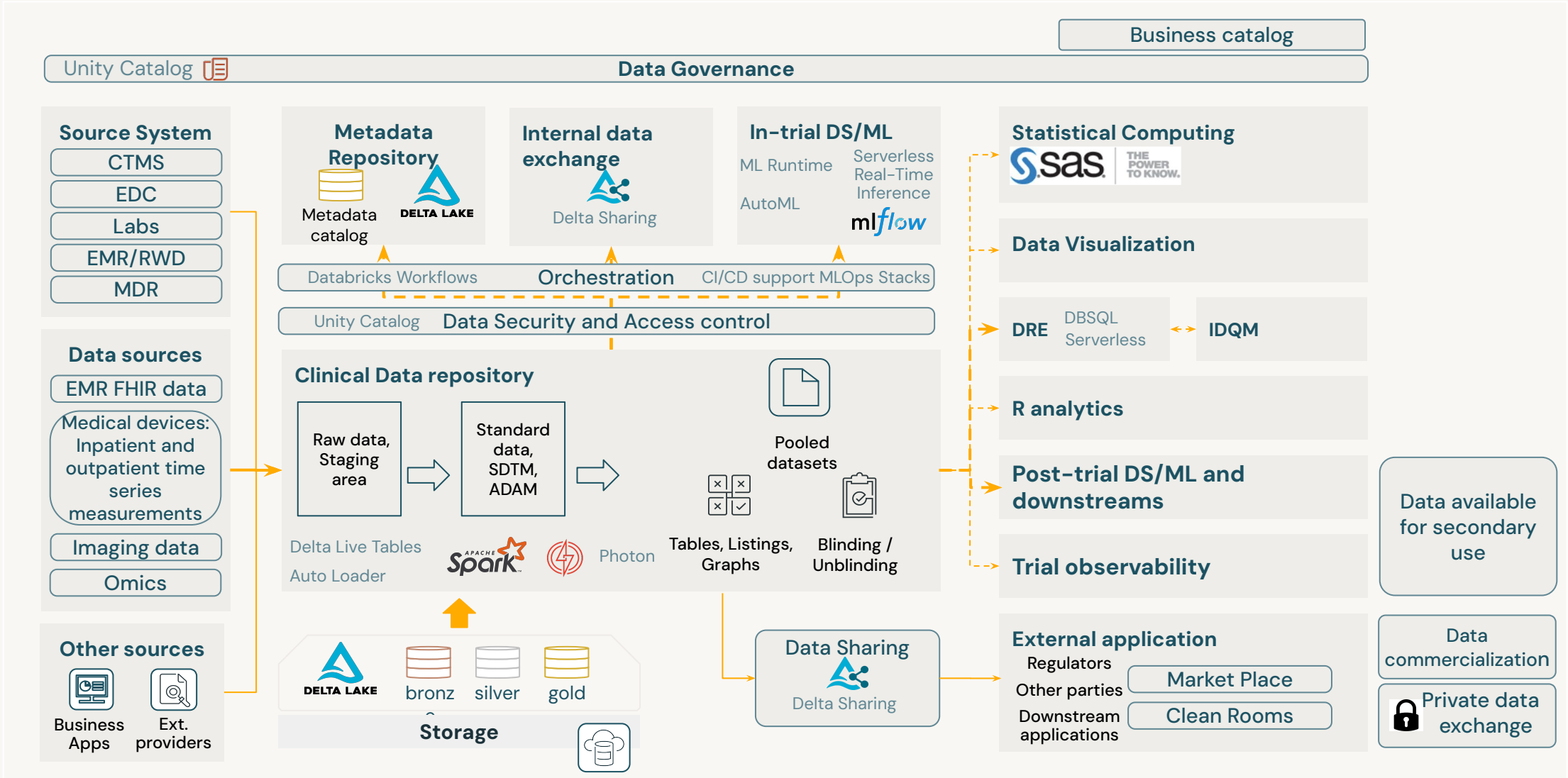
Compliance, security, and governance
Industry compliance standards like HIPAA for workspaces with security and governance at core, Unity Catalog provides out of the box governance and lineage



Data sharing and ecosystem
Consume data from HLS data vendors and monetize insights through Delta Sharing

End-to-end Lakehouse
Databricks provides end-to-end data platform with support of workflows, monitoring and alerting, data quality checks

POV: A Modern Clinical Data Platform



Proven Impact in Life Sciences

Leading global organisations trust Databricks to power their clinical data repositories and research platforms.

Organisation	Project Focus	Key Outcomes on Databricks
Novo Nordisk	FounData Platform (CDR + GenAI)	Concept to GenAI MVP in just **9 months**, democratising insights for 500+ active R&D users.
Bayer	ALYCE Platform (CDR on Azure)	**10x faster** clinical data delivery, reducing tumour response review cycles from hours to minutes.
Biogen	FAIR Platform & Genomics	Analysed 2M genomic variants in **15 minutes** (down from 2 weeks), discovering 2 new targets.
Regeneron	FAIR Data Unification	Achieved a **600x query speed** improvement, reducing ETL pipeline runtime from 3 weeks to 2 days.

12-Week Capstone Plan

Weeks 1-2

Onboarding

Sign up for Free Edition, complete Databricks training, and finalise research use case with professors.

Weeks 3-5

Data Engineering

Architect the Medallion layers and build robust PySpark/SQL data ingestion pipelines.

Weeks 6-8

Governance

Implement Unity Catalog, configure RBAC, and apply data masking to secure clinical data.

Weeks 9-12

AI/BI & Handover

Configure the AI/BI Genie semantic layer, finalise testing, and compile handover documentation.

Your Journey Starts Now

Immediate Next Steps for Week 1

Getting Started

To hit the ground running, please complete the following steps over the next few days:

1. **Sign Up:** Register for the Databricks Free Edition online to access your workspace.
2. **Get Trained:** Complete the foundational Data Engineering and Lakehouse courses via the Databricks training portal.
3. **Define Scope:** Meet with your professors to identify a specific Research project and secure your mock clinical dataset.

We will reconvene in week 3 to review your proposed architecture blueprints.

Questions?

Let's build the future of clinical research at RMIT.

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